TASK 30 — DESCRIPTIVE STATISTICS

1. Task description and objectives

Task: Calculate and interpret mean, median, mode, range, variance, standard deviation, quartiles, and percentiles. Produce statistics for at least three numerical variables and write an interpretation of the results.

Objective: Build a foundation in descriptive statistics, which is used to summarize, explore, and communicate the main characteristics of a dataset before more advanced analysis or machine learning.

Tools: Python is the programming language; Pandas loads and summarizes tabular data; NumPy supports numerical operations; Jupyter Notebook allows code to be run in small, documented cells.

2. Dataset and variables

The project uses a reproducible teaching dataset with 30 students. `Student_ID` is an identifier and should not be summarized as a measured quantity. The analysis focuses on three numeric columns:

• **Study_Hours:** daily study hours, measured in hours.

• **Attendance_Percent:** attendance, measured as a percentage.

• **Exam_Score:** examination result out of 100 points.

First inspect the data: number of rows and columns, sample records, data types, and missing values. This helps identify unsuitable columns or data-quality issues before calculations.

Student_ID	Study_Hours	Attendance_Percent	Exam_Score
S001	1.5	55	42
S002	2.0	60	48
S003	2.5	65	51
S004	3.0	70	58
S005	3.5	72	61

3. What is descriptive statistics?

Descriptive statistics are methods for organizing and summarizing observed data. They describe what is present in the dataset; they do not automatically explain why a pattern occurred or prove cause and effect. Common outputs include measures of centre (mean, median, mode), spread (range, variance, standard deviation, IQR), and relative position (quartiles and percentiles).

4. Measures of central tendency

4.1 Mean (arithmetic average)

Formula: mean = sum of all observations ÷ number of observations. The mean uses every value and is useful when values are reasonably balanced. Because it uses every value, a very high or low outlier can move the mean noticeably.

Small example: for values 2, 4, 6, the mean is (2 + 4 + 6) / 3 = 4. In Python/Pandas, use `data.mean()`.

4.2 Median

Sort values from smallest to largest. For an odd number of observations, the median is the middle value. For an even number, it is the average of the two middle values. It is more resistant to extreme observations than the mean and is often useful for skewed data such as income.

Example: values 2, 4, 100 have median 4, although the mean is 35.33. The extreme value affects the mean much more than the median. Use `data.median()`.

4.3 Mode

The mode is the most frequently occurring value. Example: 2, 3, 3, 5 has mode 3. A dataset may have multiple modes, or no repeated value. Pandas `data.mode()` can return multiple rows; `.iloc[0]` selects the first row only, so check for multiple modes when that distinction matters.

5. Measures of dispersion (spread)

5.1 Minimum and maximum

Minimum is the smallest observed value; maximum is the largest. They show the endpoints of the observed data. Use `data.min()` and `data.max()`.

5.2 Range

Formula: range = maximum – minimum. Example: if scores run from 38 to 90, range = 90 – 38 = 52. A larger range means a wider distance between the extremes, but range is sensitive to outliers and does not describe how values are distributed between the endpoints.

5.3 Variance

Variance summarizes squared distances from the mean. Population variance divides by N; sample variance divides by n–1. A sample is usually used to estimate the variability of a larger population, so this project uses Pandas `var(ddof=1)`. Squared deviations mean variance has squared units (for example, score-points²), which can be less intuitive to interpret directly.

5.4 Standard deviation

Standard deviation is the square root of variance. It expresses spread in the original units. For exam scores, a standard deviation of about 14.88 means scores commonly vary around the mean by a scale of roughly 15 points; it does not mean every score is exactly 15 points from the mean.

5.5 Quartiles and interquartile range (IQR)

Q1 is the 25th percentile; Q2 is the median (50th percentile); Q3 is the 75th percentile. IQR = Q3 – Q1 and represents the width of the middle 50% of values. IQR is less influenced by extremes than the full range. A common exploratory outlier rule flags values below Q1 – 1.5×IQR or above Q3 + 1.5×IQR; this rule flags candidates for investigation, not automatic errors.

5.6 Percentiles

A percentile describes a position in a distribution. The 10th percentile (P10) is a threshold below which approximately 10% of observations fall; the 90th percentile (P90) is a threshold below which approximately 90% fall. Percentile values can differ slightly among software because quantile interpolation methods differ. Pandas uses linear interpolation by default.

6. Pandas describe() explained

`data.describe()` is a fast summary for numeric columns. It returns count, mean, standard deviation, minimum, 25th percentile, median (50th percentile), 75th percentile, and maximum. It does not include mode, range, variance, IQR, or custom percentiles by default, so the script calculates those separately.

```
data = df[['Study_Hours', 'Attendance_Percent', 'Exam_Score']]
print(data.describe().round(2))
```

7. Full Python solution

The complete runnable program is supplied as `descriptive_statistics.py`. The core calculations are shown below. The script also checks data types and missing values, prints the results, compares mean and median, and exports a CSV summary.

```
from pathlib import Path
import pandas as pd
import numpy as np

df = pd.read_csv(Path(__file__).with_name("student_performance.csv"))
columns = ["Study_Hours", "Attendance_Percent", "Exam_Score"]
data = df[columns]

print(data.describe().round(2))

results = pd.DataFrame({
    "Count": data.count(),
    "Mean": data.mean(),
    "Median": data.median(),
    "Mode": data.mode().iloc[0],
    "Minimum": data.min(),
```

```
    "Maximum": data.max(),
    "Range": data.max() - data.min(),
    "Sample_Variance": data.var(ddof=1),
    "Sample_Std_Dev": data.std(ddof=1),
    "Q1": data.quantile(0.25),
    "Q2": data.quantile(0.50),
    "Q3": data.quantile(0.75),
    "IQR": data.quantile(0.75) - data.quantile(0.25),
    "P10": data.quantile(0.10),
    "P90": data.quantile(0.90),
}).round(2)

results.to_csv("descriptive_summary.csv", index_label="Variable")
print(results)
```

Line-by-line guide:

• `Path(__file__).with_name(...)` locates the CSV beside the script, making the project easier to move.

• `pd.read_csv(...)` loads the comma-separated file into a DataFrame.

• `df[columns]` selects only the intended numerical measurements and excludes the student identifier.

• `describe()` provides the standard quick summary; `round(2)` rounds displayed values to two decimal places.

• `mean()`, `median()`, `mode()`, `min()`, and `max()` calculate central values and endpoints.

• `max() - min()` calculates the range; `var(ddof=1)` and `std(ddof=1)` use sample formulas.

• `quantile(0.25)`, `.quantile(0.50)`, and `.quantile(0.75)` calculate Q1, Q2, and Q3.

• `Q3 - Q1` calculates the IQR; quantiles 0.10 and 0.90 calculate P10 and P90.

• `to_csv(...)` saves the results so they can be reused in a report, spreadsheet, or portfolio.

8. Results table

Variable	Mean	Median	Mode	Minimum	Maximum	Range	Sample Variance	Sample Std Dev	Q1 (25%)	Q2 (50%)	Q3 (75%)	IQR	P10	P90
Study Hours	3.93	4.00	2.00	1.00	7.00	6.00	2.74	1.65	2.62	4.00	5.00	2.38	1.95	6.05
Attendance Percent	73.93	74.50	50.00	50.00	92.00	42.00	127.58	11.30	66.50	74.50	82.75	16.25	57.90	88.10
Exam Score	64.10	64.00	38.00	38.00	90.00	52.00	221.33	14.88	52.75	64.00	75.75	23.00	43.80	84.10

9. How to interpret the results

Study Hours: mean = 3.93, median = 4.00, range = 6.00, sample standard deviation = 1.65, and IQR = 2.38. The middle half of values falls approximately between 2.62 and 5.00. Mean and median are close, so the centre is similar under both measures; this alone does not prove a perfectly symmetric distribution.

Attendance Percent: mean = 73.93, median = 74.50, range = 42.00, sample standard deviation = 11.30, and IQR = 16.25. The middle half of values falls approximately between 66.50 and 82.75. Mean and median are close, so the centre is similar under both measures; this alone does not prove a perfectly symmetric distribution.

Exam Score: mean = 64.10, median = 64.00, range = 52.00, sample standard deviation = 14.88, and IQR = 23.00. The middle half of values falls approximately between 52.75 and 75.75. Mean and median are close, so the centre is similar under both measures; this alone does not prove a perfectly symmetric distribution.

Study hours are measured in hours, attendance in percentage points, and exam score in points. Standard deviations should be interpreted within each variable's own units; comparing their numerical sizes directly can be misleading because the measurement scales differ.

10. Data quality, assumptions, and limitations

• Check for missing values before analysis; many Pandas summaries skip missing values by default, so counts can differ across columns.

• Check that numeric-looking columns are actually numeric and that units are consistent.

• An outlier may be a genuine observation or a data-entry error; investigate it rather than deleting it automatically.

• A small teaching dataset is not necessarily representative of a real student population.

• Descriptive statistics describe association and distribution; they do not establish that study hours or attendance cause better exam scores.

• Sample variance and standard deviation use n–1 (`ddof=1`). Use `ddof=0` only when calculating the population variance/standard deviation for the full population being described.

• Percentile values may vary slightly with quantile methods and interpolation choices.

11. Common mistakes to avoid

• Using the mean when strong outliers make the median more representative.

• Confusing variance with standard deviation: variance is in squared units; standard deviation is in original units.

• Saying Q1 is the smallest 25% of data. More accurately, Q1 is the 25th-percentile cutoff.

• Treating P90 as the top 90% of observations; it is the cutoff at or below which about 90% of observations lie.

• Including ID numbers in numerical calculations even though they are labels, not measurements.

• Assuming `mode()` always returns exactly one value.

• Interpreting a descriptive pattern as proof of cause and effect.

12. Interview questions and answers

Q. What is descriptive statistics?
Methods for summarizing observed data using measures such as centre, spread, and position.

Q. Difference between mean, median, and mode?
Mean is the arithmetic average, median is the middle after sorting, and mode is the most frequent value.

Q. When is median preferred over mean?
When a distribution is skewed or has extreme values, because the median is less sensitive to them.

Q. What does standard deviation tell you?
The typical scale of spread around the mean, expressed in the variable's original units.

Q. Variance vs standard deviation?
Variance is the average squared deviation; standard deviation is its square root.

Q. What is a percentile?
A distribution cutoff: about p% of observations lie at or below the pth percentile.

Q. What is IQR?
Q3 minus Q1, representing the width of the middle 50% of observations.

Q. What does describe() return?
For numeric data, count, mean, standard deviation, minimum, quartiles, and maximum.

Q. Why use ddof=1?
It uses n–1 for sample variance/standard deviation, a common estimator when data are a sample from a larger population.

Q. Can descriptive statistics prove causation?
No. They summarize observed data; causal claims need an appropriate research design and additional analysis.